**Panskura Banamali College (Autonomous)**

**B.Sc.(H) 5th Semester Examination 2022 (Online mode)**

**Subject : BCA** **Paper: CC12T**

**Paper Name : Microprocessor & Computer Organization**

**F.M. : 40** **Time : 2 Hours**

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**Answer any four question** **4×10=40**

**1.** What is stack ? Write down the operation of stack in memory. A computer uses RAM chips of 1024 × 1 capacity. How many chips are needed to provide a memory capacity of 16K bytes ? Explain in words how the chips are to be connected to the address bus.

1+ 2+ (4+3)

**2.** What are the difference between software interrupt & a subroutine call ? Explain Booth's multiplication algorithm with suitable example. Can an input port & an output port have the some port address? Justify it.

3+5+2

**3.** What is cache memory ? Describe different elements of cache design. Explain mapping.

2+5+3

4. a) What are the difference between Bit-Slice and Symbolic processors ? 2+4+4
b) Describe the Addressing Mode of 8085 .
c) What is Instructions, discuss various types of instructions in 8085?

5. a) Describe Basic Instruction cycle with diagram . 3+4+3
b) What Is A Stack Pointer Register, Describe Briefly?
c) Describe Flag Manipulation and Processor Control Instructions in 8086 .

6. Write short note : 5×2=10

(a) Program counter

(b) Miss penalty

(c) Demultiplexer

(d) Internal & external interrupt.

(e) Universal shift register

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